

LESSON 2: TYPES OF COMPUTERS

There are many types of computers. In general, computers can be classified based on their intended purpose into two broad categories: general-purpose and specific-purpose.

COMPUTERS CAN BE CLASSIFIED BASED ON THEIR PURPOSE AS
GENERAL-PURPOSE OR SPECIFIC-PURPOSE.

GENERAL PURPOSE

General-purpose computers are the most common and widely used computers. They are our personal computers: the desktop, laptop, tablet and smart-phone. These computers are designed to run a wide variety of applications which allows them to perform a wide variety of tasks that depend on the needs of the user.

So rather than being designed to perform one specific task or set of tasks, these general-purpose computers are designed to be highly flexible and to perform a wide variety of tasks. The choice of the task depends on the user and the availability of an application to perform the required task.

PERSONAL COMPUTERS ARE GENERAL-PURPOSE COMPUTERS.

EXAMPLE 2.1

Karl has a laptop that he uses for many different purposes. Here is a list of some of the tasks he can perform using the various applications he has on his laptop.

- **Writing assignments:** Using a word processing application, such as MS Word, Karl can type essays, reports or other school projects.
- **Multimedia presentations:** Using applications, such as MS PowerPoint, Karl can prepare slideshows with images and videos for his school projects.
- **Research:** Using a web browser application, such as Google Chrome, Karl can find information in the forms of articles, e-books, images and videos.
- **Collaboration:** Using applications, such as Google Drive, Karl can share his projects with his classmates so that they can complete them as a team.
- **Video Conferencing:** Using applications, such as Google Meet, Karl can communicate with his classmates to enhance collaboration on projects.
- **Gaming:** Karl has a variety of games that he can play alone or with his friends online, such as Minecraft and Fortnite.
- **Entertainment:** Using applications, such as Netflix, Karl can watch movies for entertainment.
- **Time Management:** Using applications, such as Calendar, Karl can schedule his assignments and ensure they are completed on time.

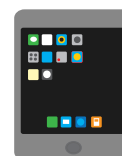
These are some of the many tasks for which Karl uses his computer. The general-purpose nature of his laptop allows him to adapt it to his various needs and preferences, which makes it a versatile and useful machine.



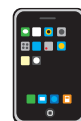
Desktop



Laptop



Tablet



Smartphone

PERSONAL COMPUTERS

Personal computers are classified into four types: desktop, laptop, tablet and smartphone computers. Flexibility is a feature common to all personal computers. Two features that distinguish these computers are portability and capability.

**PERSONAL COMPUTERS CAN BE DISTINGUISHED BASED ON
PORTABILITY AND CAPABILITY.**

Portability refers to the ease with which a computer can be carried around. This depends on three factors: size, weight and power supply. Certainly, the size and weight of a computer affect the ease with which it is carried around, making smaller and lighter computers more portable than larger and heavier ones. In addition, having a portable electrical power source, a battery, makes portability possible. Computers that need to remain connected to a power outlet remain stationary and are not portable.

**PORTABILITY DEPENDS ON THE SIZE, WEIGHT AND POWER SOURCE
OF THE COMPUTER.**

Capability refers to the complexity of tasks a computer is capable of performing. Complex tasks typically require a larger computer with a larger screen and, more importantly, greater computational power. The larger screen allows the user to clearly see the various visual elements displayed made possible by the greater computational power which allows the computer to process a lot of information and to perform many complex calculations very fast. However, computational power requires electrical power. The more information a computer processes per second, the more electrical energy it requires per second.

CAPABILITY DEPENDS MAINLY ON COMPUTATIONAL POWER.

1. **Desktops** are large, stationary computers that are designed to perform more complex tasks. Therefore, they have larger screens and greater computational power, both of which demand more electrical energy. This makes the battery an inconvenient source of electrical power as it will require frequent recharging. In addition, since complex tasks are typically performed with the user seated at a desk, a battery is not required.
2. **Laptops** are portable computers with capabilities close to those of a desktop. To achieve portability, laptops are made smaller by combining the screen and keyboard into one unit rather than being separate parts, as in desktops. In addition, laptops are powered by a rechargeable battery which can handle the energy demands of the smaller screen and the less complex tasks. Do note that certain laptops can match the capabilities of some desktops.
3. **Tablets** are not only portable; they are handheld devices that are larger than smartphones but smaller than laptops. Their smaller size is achieved through touchscreen technology which abates the need for a physical keyboard and pointing devices, such as the mouse and touchpad. They are more portable but less powerful and, therefore, less energy-demanding than laptops.
4. **Smartphones** are portable handheld devices designed primarily for communication but can also be used for a wide range of other tasks, such as browsing the internet, taking pictures and playing games. They are the most portable of the personal computers and the least capable.

In general, personal computers are designed to be usable by a wide range of people and in a wide range of settings. People of all ages, backgrounds and professions find the personal computer very useful in their lives. So not only are personal computers designed to be highly flexible, highly usable and with decent computational power, but they are also designed to deliver all that at an affordable price.

PERSONAL COMPUTERS ARE GENERALLY DESIGNED TO BALANCE CAPABILITY, FLEXIBILITY AND USABILITY WITH AFFORDABILITY.

SPECIFIC PURPOSE

In contrast to general-purpose computers, specific-purpose computers are designed to perform a specific set of functions or tasks. These computers are often specialized and designed to run specific applications rather than a broad range of applications.

SUPERCOMPUTERS

The supercomputer is a type of computer that is designed to solve complex problems. Typically, a supercomputer will handle one complex task at a time. Such a task requires massive amounts of computational power in order to handle the massive amounts of information that must be processed. Therefore, supercomputers are currently massive in size and demand immense amounts of electrical energy, making them very expensive to have and maintain.

Usually, these computers are used by major institutions, corporations and governments for the purposes of scientific research, weather forecasting, oil and gas exploration, molecular modeling and other large-scale applications.

SUPERCOMPUTERS ARE DESIGNED TO HANDLE ONE COMPLEX TASK AT A TIME.



A supercomputer focuses all its computational power on addressing one big task.

EXAMPLE 2.2

The IBM Blue Gene used by the National Oceanic and Atmospheric Administration (NOAA) in the United States is a supercomputer used for weather forecasting. The Blue Gene can perform over 1 quadrillion calculations per second, making it one of the most powerful computers in the world.

SUPERCOMPUTERS ARE OPTIMIZED FOR PERFORMANCE.

The Blue Gene can analyze vast amounts of meteorological data, including satellite images, weather station readings and oceanic data, in order to create detailed models of the atmosphere. These models can then be used to forecast weather patterns, including the trajectory of storms, the likelihood of severe weather events and long-term climate trends.

This allows for more accurate and timely predictions of severe weather events, which has important implications for disaster preparedness and response, as well as for industries such as agriculture and transportation that are heavily impacted by weather patterns.



The IBM Blue Gene/P supercomputer installation at the Argonne Leadership Angela Yang Computing Facility located in the Argonne National Laboratory in Lemont, Illinois, USA

SERVERS

The server is a computer that is designed to handle many simple tasks at the same time, rather than one complex task at a time, as would the supercomputer. The majority of these simple tasks are requests for sharing information, such as videos on the YouTube server or for processing information, such as the calculation of student averages in the database of the school server.

SERVERS ARE DESIGNED TO HANDLE MULTIPLE TASKS CONCURRENTLY.



A server focuses all its computational power on handling as many requests as it can simultaneously.

Servers vary in size depending on the workload they have to handle. Servers that are expected to receive heavy workloads are combined together, forming a data center. The combined servers are large enough to have the required computational power but require a lot of energy and are expensive. On the other hand, servers that are expected to receive light workloads tend to be smaller in size, have less computational power, require less energy and are far less expensive. Servers, therefore, vary from data centers the size of a supercomputer to a single server computer the size of a desktop.

EXAMPLE 2.3

The number of requests received by a Google data center, which houses its servers, varies depending on a range of factors, such as the time of day, the location of the data center and the specific services being used. However, it is estimated that Google processes billions of search queries each day, translating to tens of thousands of requests per second across its global network of data centers. In general, each data center may handle millions or even billions of requests per day, depending on the demand for Google's services in that particular region.



One of Google's many data centers is located in St. Ghislain, Belgium. The servers in the data center run Google products such as Search, Gmail and YouTube, as well as provide support for Google users across Europe and around the world.

EXAMPLE 2.4

Jenna's school uses a server located on-site to store and share educational resources. Jenna's teachers upload course materials, assignments and student grades to the server, which students like Jenna can access from computers on the school's network. The server also provides Jenna's parents access to her grades, attendance records and other important information about her academic progress.

Jenna's school is a medium-sized school attended by a few hundred students. For a school that size, its server would likely handle a few hundred to, at most, a few thousand requests per day. This number depends on the demand placed on the server by students, teachers and parents.



The server at Jenna's school can handle a few hundred to a few thousand requests per day, depending on the demand placed on it by the students, teachers and parents.

Servers are classified as specific-purpose devices because they are designed to perform a specific set of functions or services. Servers are labeled according to the type of function they are dedicated to performing.

Web servers	Host websites and deliver web pages
Email servers	Manage email accounts and deliver email
File servers	Store and share files
Database servers	Manage and store data in database
Game servers	Host online games and manage player accounts

Servers are dedicated to serving other computers and their users. In general, most users expect servers to respond quickly, be available any time they send a request and, most importantly, to keep their private information secure. This means that servers are designed for high performance, reliability and security.

SERVERS ARE OPTIMIZED FOR PERFORMANCE, RELIABILITY AND SECURITY.

EMBEDDED COMPUTERS

Embedded computers are computers within other devices or systems, such as cars, medical devices and home appliances and are typically hidden from view. They are classified as specific-purpose because they are designed to perform specific tasks, particularly controlling and optimizing the function of the machine in which they are embedded. In general, embedded computers are optimized for energy efficiency, small size and high reliability.

EMBEDDED COMPUTERS ARE OPTIMIZED FOR LOW POWER CONSUMPTION, SMALL SIZE AND HIGH RELIABILITY.

1. **Low power consumption:** Embedded computers are often designed to operate in environments where power may be limited, such as in remote areas or in devices that rely on battery power. To conserve power and extend battery life, embedded computers are typically designed to be energy efficient and have low power consumption.

2. **Small size:** Embedded computers are often used in small devices or appliances with limited space. This means that they must be designed to use less space and have small components that can be densely packed.
3. **High reliability:** Embedded computers are often used in devices where reliability is critical, such as in automotive or medical devices. They must be designed to operate without interruption or failure, even in harsh environments or extreme conditions. This requires components that can withstand high temperatures, shock, vibration and other stresses.

EXAMPLE 2.5

Jenna's father has a heart condition. His cardiologist informed him that his condition requires a pacemaker. The pacemaker is a device that is placed inside the body and is connected to the heart; its task is to regulate the beating of the heart.

The pacemaker has an embedded computer. The computer is tasked with reading the electrical activity of the heart and sending regular electrical pulses to maintain a steady rhythm.

Low power consumption

Since the pacemaker is placed inside the body, it must be powered by a battery. Recharging the battery requires a trip to the doctor's office, so having a long battery life is convenient.

Small size

Since the pacemaker has to fit inside the body, it should be small in size. This means its embedded computer should also be small in size.

High reliability

Since the pacemaker is responsible for keeping Jenna's father alive, it must be reliable. It must not fail. If it fails, his life will be at risk.

Overall, embedded computers are designed to meet the specific needs and enhance the functionality of the device or machine they are being used in.

SUMMARY

Computers can be classified into two broad categories: general-purpose and specific-purpose.

General-purpose computers are flexible and can run a wide range of applications and therefore perform various tasks and serve many purposes. Personal computers fall into the category of general-purpose; they are: desktops, laptops, tablets and smartphones.

Personal computers are distinguished by portability and capability. Portability depends on the size, weight and power source. Capability is largely determined by computational power. The desktop ranks the lowest in portability but the highest in capability. The smartphone ranks the highest in portability but the lowest in capability, with the laptop and tablet ranking in between. Personal computers are designed to have reasonable capability, high flexibility, easy portability and affordability.

Specific-purpose computers are designed to perform a specific set of functions or tasks. Supercomputers, servers and embedded computers are types of specific-purpose computers.

Supercomputers are massive computers that are optimized for high performance regardless of the energy consumption and the high expenses required to own, run and maintain them. They handle one complex task at a time.

Servers vary in size from data centers as large as a supercomputer to a single server computer the size of a desktop, depending on the workload they are required to handle. In all cases, these computers are optimized for high performance, high reliability and high security.

Embedded computers are computers placed inside machines and devices for specific purposes that optimize their overall function. Embedded computers are optimized for energy efficiency, small size and high reliability. They are ideal for use in devices where space and power are limited and reliability is critical.

Activity 2.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	General-purpose computers are designed to perform specific tasks.
<u>T</u>	Desktop computers are generally more powerful than laptops.
<u>T</u>	Size, weight and power supply are the three main factors that determine the portability of a computer.
<u>F</u>	Supercomputers can handle multiple complex tasks at the same time.
<u>F</u>	Servers can handle only one task at a time.
<u>F</u>	Portability is an important feature of servers.
<u>T</u>	Embedded computers are designed to be part of other devices.
<u>T</u>	Servers are typically more powerful than personal computers.
<u>T</u>	Embedded computers are optimized to be energy efficient, small in size and highly reliable.
<u>T</u>	There are different types of servers, each dedicated to a particular task.

Activity 2.2

Answer the questions below.

1. Which type of computer is designed to perform a wide range of tasks?
 - a) Specific-purpose computer
 - b) General-purpose computer
 - c) Embedded computer

2. Which of the following is an example of a specific-purpose computer?
 - a) Laptop
 - b) Desktop
 - c) Supercomputer

3. What is the primary function of a server?
 - a) To perform a wide range of tasks
 - b) To provide resources or services to other computers on a network
 - c) To control the engine of a car

4. What is the primary function of an embedded computer?
 - a) To perform a wide range of tasks
 - b) To provide resources or services to other computers on a network
 - c) To perform a specific set of functions within a larger device or system

5. Which type of computer is designed for portability and convenience?
 - a) Desktop
 - b) Laptop
 - c) Server

Activity 2.3

Answer the questions below.

1. Which type of computer is optimized for advanced scientific and engineering simulations?
 - a) Embedded computer
 - b) **Supercomputer**
 - c) Server

2. Which type of computer is designed to control the engine, monitor the fuel efficiency or manage the entertainment system in a car?
 - a) Smartphone
 - b) Server
 - c) **Embedded computer**

3. Which type of computer is best suited for managing email accounts?
 - a) Database server
 - b) **Email server**
 - c) Game server

4. Which type of computer is designed to be small and energy efficient?
 - a) Desktop
 - b) Supercomputer
 - c) **Embedded computer**

5. Which type of computer is optimized for high performance, reliability and security?
 - a) Supercomputer
 - b) **Server**
 - c) Embedded computer

Activity 2.4

Match each computer with the statement that best describes it.

	Smartphone	Desktop	Laptop
	Supercomputer	Server	Embedded computer
1.	It is a personal computer that is not portable.		<u>Desktop</u>
2.	It is massive and handles one complex task at a time.		<u>Supercomputer</u>
3.	It is small in size, highly reliable and energy efficient.		<u>Embedded computer</u>
4.	It is the smallest and most portable personal computer.		<u>Smartphone</u>
5.	It is used to provide services for other computers.		<u>Server</u>
6.	A personal computer that is portable but not handheld.		<u>Laptop</u>

Activity 2.5

Match each computer with the task for which it is best optimized.

	Desktop	Server	Smartphone
	Supercomputer	Tablet	Embedded computer
1.	Constructing a map of the visible universe		<u>Supercomputer</u>
2.	Getting real-time directions for a destination		<u>Smartphone</u>
3.	Playing an online video game		<u>Desktop</u>
4.	Controlling the doors of an elevator		<u>Embedded computer</u>
5.	Hosting websites for various clients		<u>Server</u>
6.	Drawing an illustration using a stylus		<u>Tablet</u>

Activity 2.6

Answer the questions below about personal computers.

1. Why are personal computers considered general-purpose?

Personal computers are considered to be general-purpose because they are designed to be flexible and to run a wide range of applications. This allows them to perform a wide variety of tasks, such as word processing, browsing the internet, playing games, watching videos and more.

2. “Personal computers are designed for reasonable capability, high flexibility and easy portability at an affordable price.” Is this statement true? Explain.

This statement is true. Personal computers provide the capabilities needed by the average person. Personal computers are designed to be easy to use, even for those who are not computer experts. They are flexible, with a wide range of applications available that can be used for different purposes. They are priced to be affordable for the average consumer, although the cost can vary depending on the specific features and capabilities of the device.

Activity 2.7

Answer the questions below about supercomputers and servers.

1. How would you distinguish between the supercomputer and the server?

1) Purpose: Supercomputers are designed to handle one complex task at a time, such as weather forecasting. Servers are designed to handle many simpler requests from other computers simultaneously.

2) Size: Supercomputers are typically very large and complex. Servers vary in size from large data centers to single computers the size of a desktop.

2. A security breach can happen with any computer system. Compare the impact of a security breach on a server and a supercomputer.

Security breaches are serious threats and can cause significant harm depending on the information being stolen, damaged or lost. A security breach for a server involves the theft, loss or damage to personal, medical or corporate private information stored on the server. A security breach for a supercomputer involves the theft, loss or damage to information related to national security or scientific research. This may compromise the security of the nation or the accuracy and reliability of the scientific calculations and simulations.

Activity 2.8

Answer the questions below about embedded computers. Note that they can be referred to as 'control modules', 'controllers' or 'micro-controllers', among other names.

1. List five common machines or devices in which embedded computers are found.

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- 1) Home appliances (e.g. refrigerators, washing machines)
 - 2) Medical devices (e.g. pacemakers, MRI machines)
 - 3) Automotive systems (e.g. engine control units, anti-lock brakes)
 - 4) Aircraft systems (e.g. flight control computers, navigation systems)
 - 5) Industrial equipment (e.g. elevators, assembly line machines)
 - 6) Smart home devices (e.g. thermostats, security systems)
 - 7) Wearable devices (e.g. smartwatches, fitness trackers)
 - 8) Consumer electronics (e.g. gaming consoles, digital TVs)
 - 9) ATMs and point-of-sale systems (e.g. credit card readers, cash registers)
 - 10) Traffic control systems (e.g. traffic lights, toll booths)
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2. Choose one of the machines or devices you listed in Question 1, and describe the function of its embedded computer(s).

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- 1) Engine control module (ECM) - controls the operation of the engine
 - 2) Anti-lock brake system (ABS) module - controls the brakes
 - 3) Transmission control module (TCM) - controls transmission and gear shifting
 - 4) Airbag control module (ACM) - deploys the airbag in the event of a collision
 - 5) Infotainment system - controls audio and other entertainment functions
 - 6) Climate control module - regulates the air conditioning system
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Activity 2.9

Answer the questions below about reliability. For computers, reliability refers to the ability of a computer to perform its tasks consistently without failure.

1. Compare and discuss the importance of reliability for the operation of a supercomputer used for scientific research, a server handling financial transactions and an embedded computer in an airplane. What would be the adverse consequences of failure?

Reliability is critical for the operation of any computer system, but its

importance can vary depending on the application and context.

1) For a supercomputer used for scientific research, failure could result in

delays in the research, loss of important information, loss of money spent on

the research and loss of trust in the institution conducting the research.

2) For a server handling financial transactions, failure could result in a loss of

money for customers, an opportunity for hackers to breach the system and a

loss of trust in the financial institution.

3) For an embedded computer in an airplane, failure could result in a plane

crash with loss of life of passengers and crew, loss of money as well as loss of

trust in the airlines.

Activity 2.10

Describe a typical day for yourself and the various interactions with computers you may have. Include in your example supercomputers, servers, personal computers and embedded computers.

In the morning, I wake up to the alarm I set on my smartphone (personal computer). I check if I have any messages from my friends (server). Before I get dressed, I check the weather forecast for the day (supercomputer, server). My dad drives me to school in his car (embedded computer). In class, the teacher uses the classroom desktop (personal computer) to project the day's lesson on the board. In the afternoon, I use the microwave oven (embedded computer) to make popcorn. Mom puts laundry into the washing machine (embedded computer). I use my laptop (personal computer) to sign into my school account and check my homework assignments (server). Before bed, I watch television (embedded computer).

